INCH-POUND
A-A-60035B
17 April 2024
SUPERSEDING
A-A-60035A
23 December 2022

COMMERCIAL ITEM DESCRIPTION

BOLTS, CADMIUM AND HEXAVALENT CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers bolts containing no cadmium or hexavalent-chromium in their finishes. This CID was developed to address steel fastener coatings without hazardous substances for use on military ground vehicles.

1.1 Intended use. Bolts contained herein are intended for ground vehicle applications. Bolts described in this CID shall not be used for aerospace applications without design authority approval.

1.2 Clean finish. See Section 7 (paragraph 7.3) for definition of clean finish when referenced on a table.

2 CLASSIFICATION

2.1 Types.

Type I	-	NAS1251 - Bolt, Ring, Swivel
Type II	-	NAS1297 - Bolt, Shoulder-Hexagon Head
Type III	-	NAS6603 THRU NAS6620 - Bolt, Tension, Hex Head, Close Tolerance, Alloy Steel, Long Thread, Reduced Major Dia., Self-Locking and nonlocking, 160 KSI Ftu
Type IV	-	NASM3-20 - Bolt, Machine, Aircraft
Type V	-	NASM21098 - Bolt, Self-Locking, Steel, 160 KSI Ftu, 250 °F, 12 Point, External Wrenching (Externally Wrenching Cap Screws)
Type VI	-	NASM21250 - Bolt, Tension, Steel, External Wrenching, Flanged, 12-Point, 180 KSI Ftu, 450 °F
Type VII	-	NASM35751 - Bolt, Square Neck, Round Head (Carriage), Steel, Cadmium or Zinc Plated, UNC-2A
Type VIII	-	NASM21134 - Bolt, Tension, Steel, 180 KSI Ftu, 450 °F External Wrenching, Spline Drive, Flanged
Type IX	-	NASM20073 - Bolt, Machine, Aircraft, Drilled Head, Fine Thread

3 SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, the bolts shall be manufactured as specified in the Aerospace Industries Association (AIA) documents listed above by type with the exception of, the plating/finishes shall be free of cadmium and hexavalent chromium finishes as specified herein. The use of recovered materials made in compliance with regulatory requirements is acceptable provided all requirements of this CID are met.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of the address information using the ASSIST Online database at https://quicksearch.dla.mil.

FSC 5306

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3.2 Dimensions. Dimensions for all designated type of bolts apply after plating/finishing and prior to any application of lubricants (if any). Refer to each individual NAS/NASM specification listed in Section 2.1 for detailed part geometry and tolerance requirements.

3.3 Hexavalent chromium free Part or Identifying Number (PIN) codes and finish/plating options. PIN finish/plating codes for all designated types of bolts:

D	=	Aluminum / Aluminum alloy shall be chemically conversion coated in accordance with MIL-DTL-5541, Type II, Class 1A.
N	=	Aluminum / Aluminum alloy shall be anodized in accordance with MIL-PRF-8625, Type IB, II or IIC, Class 1 or 2. Sealers shall not contain hexavalent chromium compounds.
QX	=	Carbon/alloy steel shall be Zinc Nickel plated, Type K16 in accordance with MIL-PRF-32647.
P5	=	CRES shall be cleaned, descaled and passivated in accordance with SAE AMS2700 Method 1, Type 8, or Method 2. Hexavalent chromium containing compounds shall be prohibited.
Z	=	Carbon/alloy steel shall be zinc electrodeposited plate in accordance with ASTM F1941/F1941M, Fe/Zn 5ANS.

3.4 Lubrication. Lubrication types (if any) for each type of bolt shall be specified in the referenced AIA/NASM specification (see 2.1).

3.5 Interchangeability. When AIA/NASM specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent chromium with the cadmium and hexavalent chromium free parts listed herein without design authority approval.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 14057; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this CID and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

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7 NOTES

The following part or identification numbering (PIN) procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Tables. The following finish/plating PIN's are cadmium and hexavalent chromium free for designated bolt types.

TABLE I. Type I (-1), bolt, ring, swivel.

(Use NAS1251 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	A-A-60035 Part Number Zinc Plated (Clean Finish)	NAS1251 Reference Dash Number Cadmium or Zinc Plated DO NOT USE on Ground Vehicles
AA60035-1-2*QX	AA60035-1-2*Z	NAS1251-2*
AA60035-1-5*QX	AA60035-1-5*Z	NAS1251-5*
AA60035-1-A10*QX	AA60035-1-A10*Z	NAS1251A10*
AA60035-1-A25*QX	AA60035-1-A25*Z	NAS1251A25*

NOTES:

PIN construction:

AA60035 -1 -5 * QX

- QX — Finish code (see 3.3)
  - QX = Zinc-nickel plated Type K16
  - Z = zinc plated
- * — Grip length code in 0.0625-inch rounded to three decimal places. Use "D" for drilled thread; absence of "D" means without drilled thread (see NAS1251)
- -5 — Size code (see NAS1251)
- -1 — Table/Type number in this CID (-1 = Table/Type I)
- AA60035 — CID identifier (part of basic part number)

Examples of clean finish PIN's:

AA60035-1-5D8Z = a NAS1251 - Bolt, ring, 5,000 LBF rated capacity, drilled thread, 0.500 grip length, zinc plated.

AA60035-1-A10-10QX = a NAS1251 - Bolt, ring, 10,000 LBF rated capacity, without drilled thread, 0.625-inch grip length, zinc-nickel Type K16 plated.

TABLE II. Type II (-2), bolt, shoulder-hexagon head.

(Use NAS1297 for material, heat treatment (if any), and sizes/lengths.)

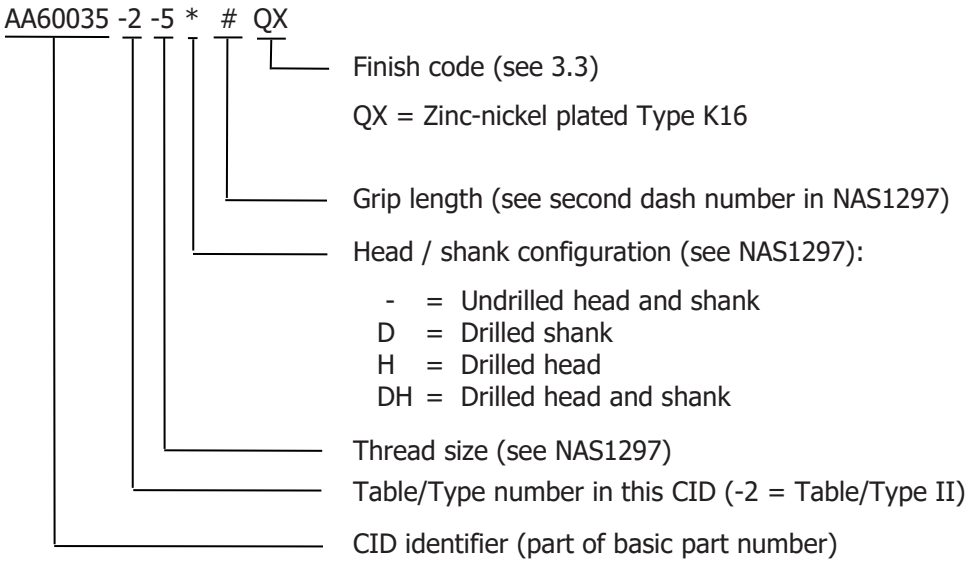
A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NAS1297 Reference Dash Number Cadmium Plated DO NOT USE on Ground Vehicles
AA60035-2-06*#QX	NAS1297-06*#
AA60035-2-3*#QX	NAS1297-3*#
AA60035-2-4*#QX	NAS1297-4*#
AA60035-2-5*#QX	NAS1297-5*#
AA60035-2-6*#QX	NAS1297-6*#
AA60035-2-7*#QX	NAS1297-7*#
AA60035-2-8*#QX	NAS1297-8*#
AA60035-2-10*#QX	NAS1297-10*#

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NOTES:

PIN construction:



Examples of clean finish PIN's:

AA60035-2-06-4QX = a NAS1297 - Shoulder bolt with a hexagon head, alloy steel, 0.1380-32 UNJC-3A, undrilled, 0.260 grip length, zinc-nickel Type K16 plated.

AA60035-2-5D8QX = a NAS1297 - Shoulder bolt with a hexagon head, alloy steel, 0.3125-24 UNJF-3A, drilled shank, 0.510 grip length, zinc-nickel Type K16 plated.

AA60035-2-5H8QX = a NAS1297 - Shoulder bolt with a hexagon head, alloy steel, 0.3125-24 UNJF-3A, drilled head, 0.510 grip length, zinc-nickel Type K16 plated.

AA60035-2-10DH10QX = a NAS1297 - Shoulder bolt with a hexagon head, alloy steel, 0.6250-18 UNJF-3A, drilled shank and head, 0.635 grip length, zinc-nickel Type K16 plated.

TABLE III. Type III (-3), bolt, tension, hex head, close tolerance, alloy steel, long thread, reduced major dia., self-locking and nonlocking, 160 KSI Ftu.

(Use NAS6603 THRU NAS6620 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NAS6603 THRU NAS6620 Reference Dash Number Cadmium Plated DO NOT USE on Ground Vehicles
AA60035-3-03*QX	NAS6603*
AA60035-3-04*QX	NAS6604*
AA60035-3-05*QX	NAS6605*
AA60035-3-06*QX	NAS6606*
AA60035-3-07*QX	NAS6607*
AA60035-3-08*QX	NAS6608*
AA60035-3-09*QX	NAS6609*
AA60035-3-10*QX	NAS6610*
AA60035-3-12*QX	NAS6612*
AA60035-3-14*QX	NAS6614*
AA60035-3-16*QX	NAS6616*
AA60035-3-18*QX	NAS6618*
AA60035-3-20*QX	NAS6620*

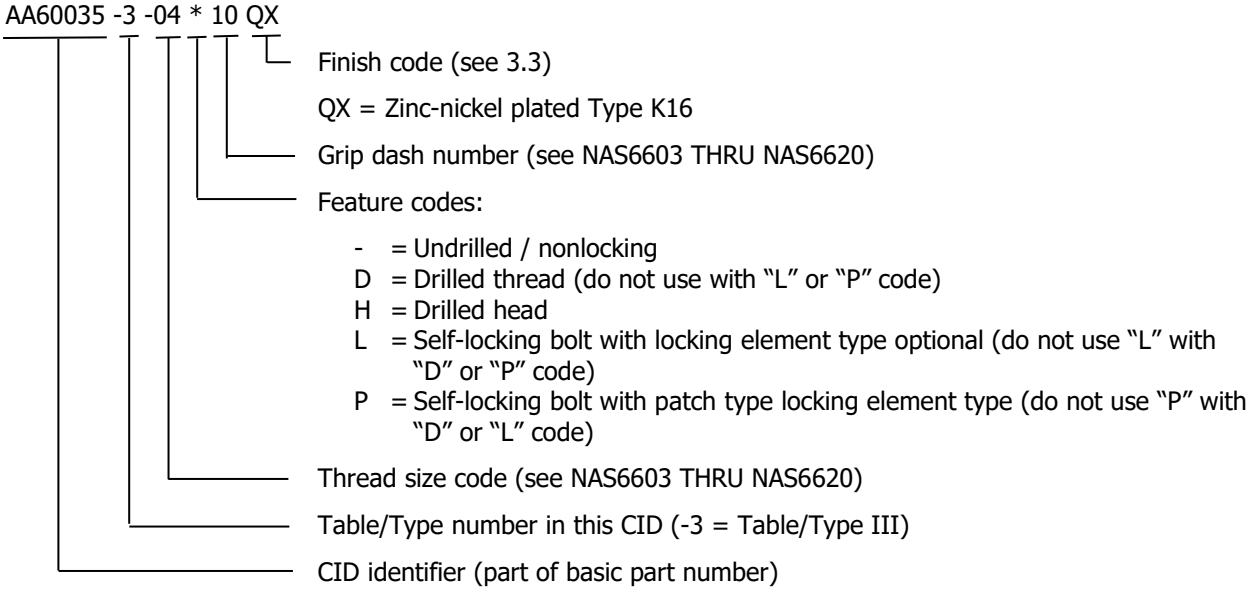
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NOTES:

(For oversized bolts, see requirements of NAS6603 THRU NAS6620, however, use finish specified below.)

PIN construction:



Examples of clean finish PIN's:

AA60035-3-04-10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, undrilled / nonlocking, zinc-nickel Type K16 plated.

AA60035-3-04D10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, drilled thread, undrilled head, nonlocking, zinc-nickel Type K16 plated.

AA60035-3-04DH10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, drilled thread, drilled head, nonlocking, zinc-nickel Type K16 plated.

AA60035-3-04H10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, undrilled thread, drilled head, nonlocking, zinc-nickel Type K16 plated.

AA60035-3-04L10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, self-locking (locking type optional), zinc-nickel Type K16 plated.

AA60035-3-04P10QX = a NAS6604 - Bolt, 0.2500-28 thread, 0.625 grip, patch type locking element type, zinc-nickel Type K16 plated.

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TABLE IV. Type IV (-4) - bolt, machine, aircraft.

(Use NASM3-20 for material, heat treatment (if any), and sizes/lengths.)
(NASM6812 is the procurement specification for NASM3-20.)

A-A-60035 Part Number Steel (Clean Finish)	NASM3-20 Reference Dash Number Cadmium Plated , DO NOT USE on Ground Vehicles	A-A-60035 Part Number CRES Passivated (Clean Finish)	NASM3-20 Reference Dash Number Passivated, (Contains Hazardous Material) DO NOT USE on Ground Vehicles	A-A-60035 Part Number Aluminum Anodized or Chemically Treated (Clean Finish)	NASM3-20 Reference Dash Number Aluminum Anodized or Chemically Treated, (Contains Hazardous Material) DO NOT USE on Ground Vehicles
AA60035-4-3-*	AN3-**	AA60035-4-3C*	AN3C**	AA60035-4-3DD*	AN3DD**
AA60035-4-4-*	AN4-**	AA60035-4-4C*	AN4C**	AA60035-4-4DD*	AN4DD**
AA60035-4-5-*	AN5-**	AA60035-4-5C*	AN5C**	AA60035-4-5DD*	AN5DD**
AA60035-4-6-*	AN6-**	AA60035-4-6C*	AN6C**	AA60035-4-6DD*	AN6DD**
AA60035-4-7-*	AN7-**	AA60035-4-7C*	AN7C**	AA60035-4-7DD*	AN7DD**
AA60035-4-8-*	AN8-**	AA60035-4-8C*	AN8C**	AA60035-4-8DD*	AN8DD**
AA60035-4-9-*	AN9-**	AA60035-4-9C*	AN9C**	AA60035-4-9DD*	AN9DD**
AA60035-4-10-*	AN10-**	AA60035-4-10C*	AN10C**	AA60035-4-10DD*	AN10DD**
AA60035-4-12-*	AN12-**	AA60035-4-12C*	AN12C**	AA60035-4-12DD*	AN12DD**
AA60035-4-14-*	AN14-**	AA60035-4-14C*	AN14C**	AA60035-4-14DD*	AN14DD**
AA60035-4-16-*	AN16-**	AA60035-4-16C*	AN16C**	AA60035-4-16DD*	AN16DD**
AA60035-4-17-*	AN17-**	AA60035-4-17C*	AN17C**	AA60035-4-17DD*	AN17DD**
AA60035-4-18-*	AN18-**	AA60035-4-18C*	AN18C**	AA60035-4-18DD*	AN18DD**
AA60035-4-20-*	AN20-**	AA60035-4-20C*	AN20C**	AA60035-4-20DD*	AN20DD**

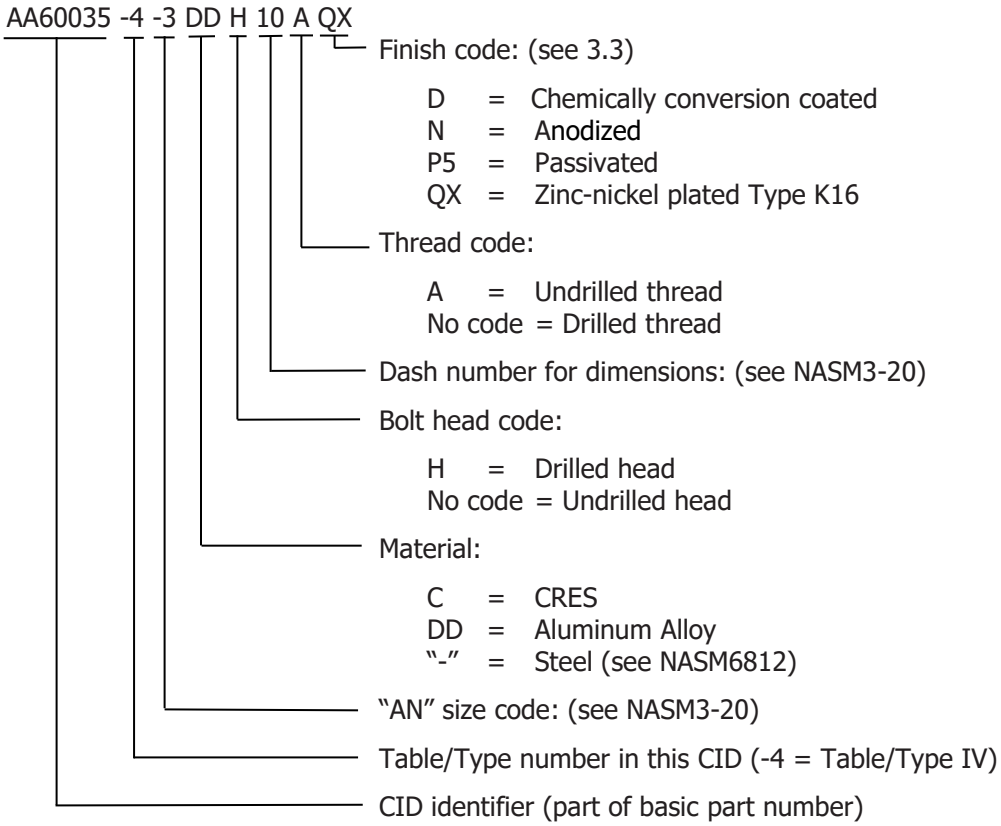
NOTES:

* See PIN construction for part number breakdown.

** See NASM3-20 for part number breakdown.

(AA60035-4-16 is inactive for new design after June 15, 1966.)

PIN construction:



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Examples of clean finish PIN's:

AA60035-4-6-10QX = a NASM3-20 - Ø.3750 steel bolt 1.078 long, 0.438 grip with drilled threads only, zinc-nickel Type K16 plated.
AA60035-4-6C10P5 = a NASM3-20 - Ø.3750 CRES bolt 1.078 long, 0.438 grip with drilled threads only, passivated.
AA60035-4-6DD10D = a NASM3-20 - Ø.3750 aluminum alloy bolt 1.078 long, 0.438 grip with drilled threads only, chemically conversion coated.
AA60035-4-6DDH10D = a NASM3-20 - Ø.3750 aluminum alloy bolt 1.078 long, 0.438 grip with drilled head and threads, chemically conversion coated.
AA60035-4-6DD10AN = a NASM3-20 - Ø.3750 aluminum alloy bolt 1.078 long, 0.438 grip, undrilled threads and head, anodized.

TABLE V. Type V (-5), bolt, self-locking, steel, 160 KSI Ftu, 250 °F, 12 point, external wrenching (externally wrenching cap screws).

(Use NASM21098 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NASM21098 Part/Dash Number Cadmium Plated, DO NOT USE on Ground Vehicles	A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NASM21098 Part/Dash Number Cadmium Plated, DO NOT USE on Ground Vehicles
AA60035-5-17QX	MS21098-17	AA60035-5-73QX	MS21098-73
AA60035-5-18QX	MS21098-18	AA60035-5-74QX	MS21098-74
AA60035-5-23QX	MS21098-23	AA60035-5-75QX	MS21098-75
AA60035-5-24QX	MS21098-24	AA60035-5-76QX	MS21098-76
AA60035-5-25QX	MS21098-25	AA60035-5-77QX	MS21098-77
AA60035-5-30QX	MS21098-30	AA60035-5-79QX	MS21098-79
AA60035-5-31QX	MS21098-31	AA60035-5-80QX	MS21098-80
AA60035-5-32QX	MS21098-32	AA60035-5-81QX	MS21098-81
AA60035-5-37QX	MS21098-37	AA60035-5-82QX	MS21098-82
AA60035-5-38QX	MS21098-38	AA60035-5-83QX	MS21098-83
AA60035-5-39QX	MS21098-39	AA60035-5-84QX	MS21098-84
AA60035-5-44QX	MS21098-44	AA60035-5-85QX	MS21098-85
AA60035-5-45QX	MS21098-45	AA60035-5-86QX	MS21098-86
AA60035-5-46QX	MS21098-46	AA60035-5-87QX	MS21098-87
AA60035-5-50QX	MS21098-50	AA60035-5-88QX	MS21098-88
AA60035-5-51QX	MS21098-51	AA60035-5-89QX	MS21098-89
AA60035-5-52QX	MS21098-52	AA60035-5-90QX	MS21098-90
AA60035-5-56QX	MS21098-56	AA60035-5-91QX	MS21098-91
AA60035-5-57QX	MS21098-57	AA60035-5-92QX	MS21098-92
AA60035-5-58QX	MS21098-58	AA60035-5-93QX	MS21098-93
AA60035-5-59QX	MS21098-59	AA60035-5-94QX	MS21098-94
AA60035-5-60QX	MS21098-60	AA60035-5-95QX	MS21098-95
AA60035-5-61QX	MS21098-61	AA60035-5-96QX	MS21098-96
AA60035-5-64QX	MS21098-64	AA60035-5-97QX	MS21098-97
AA60035-5-65QX	MS21098-65	AA60035-5-98QX	MS21098-98
AA60035-5-66QX	MS21098-66	AA60035-5-99QX	MS21098-99
AA60035-5-67QX	MS21098-67	AA60035-5-100QX	MS21098-100
AA60035-5-68QX	MS21098-68	AA60035-5-101QX	MS21098-101
AA60035-5-69QX	MS21098-69	AA60035-5-102QX	MS21098-102
AA60035-5-72QX	MS21098-72		

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NOTES:

PIN construction

AA60035 -5 -58 QX

- QX = Finish code (see 3.3)
- QX = Zinc-nickel plated Type K16
- -58 = Dash number (see NASM21098 for dimensions)
- -5 = Table/Type number in this CID (-5 = Table/Type V)
- AA60035 = CID identifier (part of basic part number)

Examples of clean finish PIN's:

AA60035-5-58QX = a NASM21098 – bolt, self-locking, 12 point, steel, 0.3750-24 UNF-3A thread, 1 inch long, zinc-nickel Type K16 plated.

AA60035-5-100QX = a NASM21098 – bolt, self-locking, 12 point, steel, 0.5000-20 UNF-3A thread, 3 inch long, zinc-nickel Type K16 plated.

TABLE VI. Type VI (-6), bolt, tension, steel, external wrenching, flanged, 12-Point, 180 KSI Ftu, 450 ºF.

(Use NASM21250 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NASM21250 Reference Dash Number Cadmium Plated DO NOT USE on Ground Vehicles
AA60035-6-02*QX	MS21250-02*
AA60035-6-03*QX	MS21250-03*
AA60035-6-04*QX	MS21250-04*
AA60035-6-05*QX	MS21250-05*
AA60035-6-06*QX	MS21250-06*
AA60035-6-07*QX	MS21250-07*
AA60035-6-08*QX	MS21250-08*
AA60035-6-09*QX	MS21250-09*
AA60035-6-10*QX	MS21250-10*
AA60035-6-12*QX	MS21250-12*
AA60035-6-14*QX	MS21250-14*
AA60035-6-16*QX	MS21250-16*
AA60035-6-18*QX	MS21250-18*
AA60035-6-20*QX	MS21250-20*
AA60035-6-22*QX	MS21250-22*
AA60035-6-24*QX	MS21250-24*

NOTES:

* Grip dash number from NASM21250.

PIN construction

AA60035 -6 - 04 008 QX

- QX = Finish code (see 3.3)
- QX = Zinc-nickel plated Type K16
- 008 = Grip dash number in sixteenths of an inch (see NASM21250).
- 04 = Diameter (thread) dash number, add letter "H" before diameter dash number to designate drilled head (see NASM21250)
- -6 = Table/Type number in this CID (-6 = Table/Type VI)
- AA60035 = CID identifier (part of basic part number)

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Examples of clean finish PIN's:

AA60035-6-02008QX = a NASM21250 – Bolt undrilled head, 0.1640-32 UNJC-3A, 0.500-inch grip length, zinc-nickel Type K16 plated.
AA60035-6-04008QX = a NASM21250 – Bolt undrilled head, 0.2500-28 UNJF-3A, 0.500-inch grip length, zinc-nickel Type K16 plated.
AA60035-6-H04008QX = a NASM21250 – Bolt drilled head, 0.2500-28 UNJF-3A, 0.500 grip length, zinc-nickel Type K16 plated.

TABLE VII. Type VII (-7), bolt, square neck, round head (carriage), steel, UNC-2A.

(Use NASM35751 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	A-A-60035 Part Number Zinc Plated (Clean Finish)	NASM35751 Reference Part/Dash Number Cadmium Plated or Zinc (with Hex-Chrome) Coated, DO NOT USE on Ground Vehicles
AA60035-7-1QX	AA60035-7-1Z	MS35751-1
AA60035-7-2QX	AA60035-7-2Z	MS35751-2
AA60035-7-3QX	AA60035-7-3Z	MS35751-3
AA60035-7-4QX	AA60035-7-4Z	MS35751-4
AA60035-7-5QX	AA60035-7-5Z	MS35751-5
AA60035-7-6QX	AA60035-7-6Z	MS35751-6
AA60035-7-7QX	AA60035-7-7Z	MS35751-7
AA60035-7-8QX	AA60035-7-8Z	MS35751-8
AA60035-7-9QX	AA60035-7-9Z	MS35751-9
AA60035-7-10QX	AA60035-7-10Z	MS35751-10
AA60035-7-11QX	AA60035-7-11Z	MS35751-11
AA60035-7-12QX	AA60035-7-12Z	MS35751-12
AA60035-7-13QX	AA60035-7-13Z	MS35751-13
AA60035-7-14QX	AA60035-7-14Z	MS35751-14
AA60035-7-15QX	AA60035-7-15Z	MS35751-15
AA60035-7-16QX	AA60035-7-16Z	MS35751-16
AA60035-7-17QX	AA60035-7-17Z	MS35751-17
AA60035-7-18QX	AA60035-7-18Z	MS35751-18
AA60035-7-19QX	AA60035-7-19Z	MS35751-19
AA60035-7-20QX	AA60035-7-20Z	MS35751-20
AA60035-7-21QX	AA60035-7-21Z	MS35751-21
AA60035-7-22QX	AA60035-7-22Z	MS35751-22
AA60035-7-23QX	AA60035-7-23Z	MS35751-23
AA60035-7-24QX	AA60035-7-24Z	MS35751-24
AA60035-7-25QX	AA60035-7-25Z	MS35751-25
AA60035-7-26QX	AA60035-7-26Z	MS35751-26
AA60035-7-27QX	AA60035-7-27Z	MS35751-27
AA60035-7-28QX	AA60035-7-28Z	MS35751-28
AA60035-7-29QX	AA60035-7-29Z	MS35751-29
AA60035-7-30QX	AA60035-7-30Z	MS35751-30
AA60035-7-31QX	AA60035-7-31Z	MS35751-31
AA60035-7-32QX	AA60035-7-32Z	MS35751-32
AA60035-7-33QX	AA60035-7-33Z	MS35751-33
AA60035-7-34QX	AA60035-7-34Z	MS35751-34
AA60035-7-35QX	AA60035-7-35Z	MS35751-35
AA60035-7-36QX	AA60035-7-36Z	MS35751-36
AA60035-7-40QX	AA60035-7-40Z	MS35751-40
AA60035-7-41QX	AA60035-7-41Z	MS35751-41
AA60035-7-42QX	AA60035-7-42Z	MS35751-42
AA60035-7-43QX	AA60035-7-43Z	MS35751-43
AA60035-7-44QX	AA60035-7-44Z	MS35751-44
AA60035-7-45QX	AA60035-7-45Z	MS35751-45

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TABLE VII. Type VII (-7), bolt, square neck, round head (carriage), steel, UNC-2A- Continued.

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	A-A-60035 Part Number Zinc Plated (Clean Finish)	NASM35751 Reference Part/Dash Number Cadmium Plated or Zinc (with Hex-Chrome) Coated, DO NOT USE on Ground Vehicles
AA60035-7-46QX	AA60035-7-46Z	MS35751-46
AA60035-7-47QX	AA60035-7-47Z	MS35751-47
AA60035-7-48QX	AA60035-7-48Z	MS35751-48
AA60035-7-49QX	AA60035-7-49Z	MS35751-49
AA60035-7-50QX	AA60035-7-50Z	MS35751-50
AA60035-7-51QX	AA60035-7-51Z	MS35751-51
AA60035-7-52QX	AA60035-7-52Z	MS35751-52
AA60035-7-53QX	AA60035-7-53Z	MS35751-53
AA60035-7-54QX	AA60035-7-54Z	MS35751-54
AA60035-7-55QX	AA60035-7-55Z	MS35751-55
AA60035-7-56QX	AA60035-7-56Z	MS35751-56
AA60035-7-57QX	AA60035-7-57Z	MS35751-57
AA60035-7-58QX	AA60035-7-58Z	MS35751-58
AA60035-7-59QX	AA60035-7-59Z	MS35751-59
AA60035-7-60QX	AA60035-7-60Z	MS35751-60
AA60035-7-61QX	AA60035-7-61Z	MS35751-61
AA60035-7-63QX	AA60035-7-63Z	MS35751-63
AA60035-7-64QX	AA60035-7-64Z	MS35751-64
AA60035-7-65QX	AA60035-7-65Z	MS35751-65
AA60035-7-68QX	AA60035-7-68Z	MS35751-68
AA60035-7-69QX	AA60035-7-69Z	MS35751-69
AA60035-7-70QX	AA60035-7-70Z	MS35751-70
AA60035-7-71QX	AA60035-7-71Z	MS35751-71
AA60035-7-72QX	AA60035-7-72Z	MS35751-72
AA60035-7-73QX	AA60035-7-73Z	MS35751-73
AA60035-7-74QX	AA60035-7-74Z	MS35751-74
AA60035-7-75QX	AA60035-7-75Z	MS35751-75
AA60035-7-76QX	AA60035-7-76Z	MS35751-76
AA60035-7-77QX	AA60035-7-77Z	MS35751-77
AA60035-7-78QX	AA60035-7-78Z	MS35751-78
AA60035-7-79QX	AA60035-7-79Z	MS35751-79
AA60035-7-80QX	AA60035-7-80Z	MS35751-80
AA60035-7-81QX	AA60035-7-81Z	MS35751-81
AA60035-7-82QX	AA60035-7-82Z	MS35751-82
AA60035-7-83QX	AA60035-7-83Z	MS35751-83
AA60035-7-84QX	AA60035-7-84Z	MS35751-84
AA60035-7-85QX	AA60035-7-85Z	MS35751-85
AA60035-7-86QX	AA60035-7-86Z	MS35751-86
AA60035-7-87QX	AA60035-7-87Z	MS35751-87
AA60035-7-88QX	AA60035-7-88Z	MS35751-88
AA60035-7-89QX	AA60035-7-89Z	MS35751-89
AA60035-7-90QX	AA60035-7-90Z	MS35751-90
AA60035-7-91QX	AA60035-7-91Z	MS35751-91
AA60035-7-92QX	AA60035-7-92Z	MS35751-92
AA60035-7-93QX	AA60035-7-93Z	MS35751-93
AA60035-7-94QX	AA60035-7-94Z	MS35751-94
AA60035-7-95QX	AA60035-7-95Z	MS35751-95
AA60035-7-97QX	AA60035-7-97Z	MS35751-97
AA60035-7-98QX	AA60035-7-98Z	MS35751-98
AA60035-7-99QX	AA60035-7-99Z	MS35751-99
AA60035-7-100QX	AA60035-7-100Z	MS35751-100
AA60035-7-101QX	AA60035-7-101Z	MS35751-101
AA60035-7-102QX	AA60035-7-102Z	MS35751-102

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TABLE VII. Type VII (-7), bolt, square neck, round head (carriage), steel, UNC-2A- Continued.

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	A-A-60035 Part Number Zinc Plated (Clean Finish)	NASM35751 Reference Part/Dash Number Cadmium Plated or Zinc (with Hex-Chrome) Coated, DO NOT USE on Ground Vehicles
AA60035-7-103QX	AA60035-7-103Z	MS35751-103
AA60035-7-104QX	AA60035-7-104Z	MS35751-104
AA60035-7-105QX	AA60035-7-105Z	MS35751-105
AA60035-7-106QX	AA60035-7-106Z	MS35751-106
AA60035-7-107QX	AA60035-7-107Z	MS35751-107
AA60035-7-108QX	AA60035-7-108Z	MS35751-108
AA60035-7-109QX	AA60035-7-109Z	MS35751-109
AA60035-7-110QX	AA60035-7-110Z	MS35751-110
AA60035-7-111QX	AA60035-7-111Z	MS35751-111
AA60035-7-112QX	AA60035-7-112Z	MS35751-112
AA60035-7-113QX	AA60035-7-113Z	MS35751-113
AA60035-7-114QX	AA60035-7-114Z	MS35751-114
AA60035-7-115QX	AA60035-7-115Z	MS35751-115
AA60035-7-116QX	AA60035-7-116Z	MS35751-116
AA60035-7-117QX	AA60035-7-117Z	MS35751-117
AA60035-7-119QX	AA60035-7-119Z	MS35751-119
AA60035-7-121QX	AA60035-7-121Z	MS35751-121
AA60035-7-123QX	AA60035-7-123Z	MS35751-123
AA60035-7-124QX	AA60035-7-124Z	MS35751-124
AA60035-7-125QX	AA60035-7-125Z	MS35751-125
AA60035-7-126QX	AA60035-7-126Z	MS35751-126
AA60035-7-127QX	AA60035-7-127Z	MS35751-127
AA60035-7-128QX	AA60035-7-128Z	MS35751-128
AA60035-7-129QX	AA60035-7-129Z	MS35751-129
AA60035-7-130QX	AA60035-7-130Z	MS35751-130
AA60035-7-132QX	AA60035-7-132Z	MS35751-132
AA60035-7-133QX	AA60035-7-133Z	MS35751-133
AA60035-7-134QX	AA60035-7-134Z	MS35751-134
AA60035-7-135QX	AA60035-7-135Z	MS35751-135
AA60035-7-136QX	AA60035-7-136Z	MS35751-136
AA60035-7-137QX	AA60035-7-137Z	MS35751-137
AA60035-7-138QX	AA60035-7-138Z	MS35751-138
AA60035-7-139QX	AA60035-7-139Z	MS35751-139
AA60035-7-140QX	AA60035-7-140Z	MS35751-140
AA60035-7-141QX	AA60035-7-141Z	MS35751-141
AA60035-7-142QX	AA60035-7-142Z	MS35751-142
AA60035-7-143QX	AA60035-7-143Z	MS35751-143
AA60035-7-144QX	AA60035-7-144Z	MS35751-144
AA60035-7-145QX	AA60035-7-145Z	MS35751-145
AA60035-7-146QX	AA60035-7-146Z	MS35751-146
AA60035-7-147QX	AA60035-7-147Z	MS35751-147
AA60035-7-148QX	AA60035-7-148Z	MS35751-148
AA60035-7-149QX	AA60035-7-149Z	MS35751-149
AA60035-7-152QX	AA60035-7-152Z	MS35751-152
AA60035-7-153QX	AA60035-7-153Z	MS35751-153
AA60035-7-154QX	AA60035-7-154Z	MS35751-154
AA60035-7-156QX	AA60035-7-156Z	MS35751-156
AA60035-7-157QX	AA60035-7-157Z	MS35751-157
AA60035-7-158QX	AA60035-7-158Z	MS35751-158
AA60035-7-159QX	AA60035-7-159Z	MS35751-159
AA60035-7-160QX	AA60035-7-160Z	MS35751-160
AA60035-7-162QX	AA60035-7-162Z	MS35751-162

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TABLE VII. Type VII (-7), bolt, square neck, round head (carriage), steel, UNC-2A- Continued.

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	A-A-60035 Part Number Zinc Plated (Clean Finish)	NASM35751 Reference Part/Dash Number Cadmium Plated or Zinc (with Hex-Chrome) Coated, DO NOT USE on Ground Vehicles
AA60035-7-163QX	AA60035-7-163Z	MS35751-163
AA60035-7-164QX	AA60035-7-164Z	MS35751-164
AA60035-7-165QX	AA60035-7-165Z	MS35751-165
AA60035-7-166QX	AA60035-7-166Z	MS35751-166
AA60035-7-167QX	AA60035-7-167Z	MS35751-167
AA60035-7-168QX	AA60035-7-168Z	MS35751-168
AA60035-7-169QX	AA60035-7-169Z	MS35751-169
AA60035-7-170QX	AA60035-7-170Z	MS35751-170
AA60035-7-172QX	AA60035-7-172Z	MS35751-172
AA60035-7-174QX	AA60035-7-174Z	MS35751-174
AA60035-7-175QX	AA60035-7-175Z	MS35751-175
AA60035-7-176QX	AA60035-7-176Z	MS35751-176
AA60035-7-181QX	AA60035-7-181Z	MS35751-181
AA60035-7-182QX	AA60035-7-182Z	MS35751-182
AA60035-7-183QX	AA60035-7-183Z	MS35751-183
AA60035-7-185QX	AA60035-7-185Z	MS35751-185
AA60035-7-187QX	AA60035-7-187Z	MS35751-187
AA60035-7-189QX	AA60035-7-189Z	MS35751-189
AA60035-7-190QX	AA60035-7-190Z	MS35751-190
AA60035-7-191QX	AA60035-7-191Z	MS35751-191
AA60035-7-192QX	AA60035-7-192Z	MS35751-192
AA60035-7-193QX	AA60035-7-193Z	MS35751-193
AA60035-7-194QX	AA60035-7-194Z	MS35751-194
AA60035-7-195QX	AA60035-7-195Z	MS35751-195
AA60035-7-196QX	AA60035-7-196Z	MS35751-196
AA60035-7-197QX	AA60035-7-197Z	MS35751-197
AA60035-7-198QX	AA60035-7-198Z	MS35751-198
AA60035-7-199QX	AA60035-7-199Z	MS35751-199
AA60035-7-201QX	AA60035-7-201Z	MS35751-201
AA60035-7-202QX	AA60035-7-202Z	MS35751-202
AA60035-7-203QX	AA60035-7-203Z	MS35751-203
AA60035-7-204QX	AA60035-7-204Z	MS35751-204
AA60035-7-205QX	AA60035-7-205Z	MS35751-205
AA60035-7-206QX	AA60035-7-206Z	MS35751-206
AA60035-7-207QX	AA60035-7-207Z	MS35751-207
AA60035-7-208QX	AA60035-7-208Z	MS35751-208
AA60035-7-209QX	AA60035-7-209Z	MS35751-209
AA60035-7-210QX	AA60035-7-210Z	MS35751-210
AA60035-7-211QX	AA60035-7-211Z	MS35751-211
AA60035-7-212QX	AA60035-7-212Z	MS35751-212
AA60035-7-213QX	AA60035-7-213Z	MS35751-213
AA60035-7-214QX	AA60035-7-214Z	MS35751-214
AA60035-7-215QX	AA60035-7-215Z	MS35751-215
AA60035-7-216QX	AA60035-7-216Z	MS35751-216
AA60035-7-217QX	AA60035-7-217Z	MS35751-217
AA60035-7-218QX	AA60035-7-218Z	MS35751-218

(AA60035-7-97 through AA60035-7-121, and AA60035-7-207 and AA60035-7-208 are inactive for new design.)

(Notes continued next page.)

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NOTES:

PIN construction

AA60035 -7 - * QX

- QX = Finish code (see 3.3)
  - QX = Zinc-nickel plated Type K16
  - Z = Zinc plated
- Dash number (dimensions) (see NASM35751).
- Table/Type number in this CID (-7 = Table/Type VII)
- CID identifier (part of basic part number)

Examples of clean finish PIN's:

AA60035-7-1QX = a NASM35751 – Bolt, square neck, round head (carriage), steel, 0.1900-24 UNC-2A thread, 0.750 length, zinc-nickel Type K16 plated.

AA60035-7-81Z = a NASM35751 – Bolt, square neck, round head (carriage), steel, 0.3750-16 UNC-2A thread, 4.000 length, zinc plated.

TABLE VIII. Type VIII (-8), bolt, tension, steel, 180 KSI Ftu, 450 °F external wrenching, spline drive, flanged.

(Use NASM21134 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NASM21134 Reference Part/Dash Number Cadmium Plated, DO NOT USE on Ground Vehicles
AA60035-8-03*QX	MS21134-03*
AA60035-8-04*QX	MS21134-04*
AA60035-8-05*QX	MS21134-05*
AA60035-8-06*QX	MS21134-06*
AA60035-8-07*QX	MS21134-07*
AA60035-8-08*QX	MS21134-08*
AA60035-8-09*QX	MS21134-09*
AA60035-8-10*QX	MS21134-10*
AA60035-8-12*QX	MS21134-12*
AA60035-8-14*QX	MS21134-14*
AA60035-8-16*QX	MS21134-16*
AA60035-8-18*QX	MS21134-18*
AA60035-8-20*QX	MS21134-20*
AA60035-8-22*QX	MS21134-22*
AA60035-8-24*QX	MS21134-24*

NOTES:

PIN construction:

AA60035 -8 - 04 * QX

- QX = Finish code (see 3.3)
  - QX = Zinc-nickel plated Type K16
- Grip Length dash number (see NASM21134).
- Diameter dash number (see NASM21134).
- Head style: "-" = Undrilled, "H" = Drilled
- Table/Type number in this CID (-8 = Table/Type VIII)
- CID identifier (part of basic part number)

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Example of clean finish PIN's:

AA60035-8-04008QX = a NASM21134 - bolt, 0.2500-28 bolt, 0.970 long, 0.500 grip, undrilled head, zinc-nickel Type K16 plated.
AA60035-8H04008QX = a NASM21134 - bolt, 0.2500-28 bolt, 0.970 long, 0.500 grip, drilled head, zinc-nickel Type K16 plated.

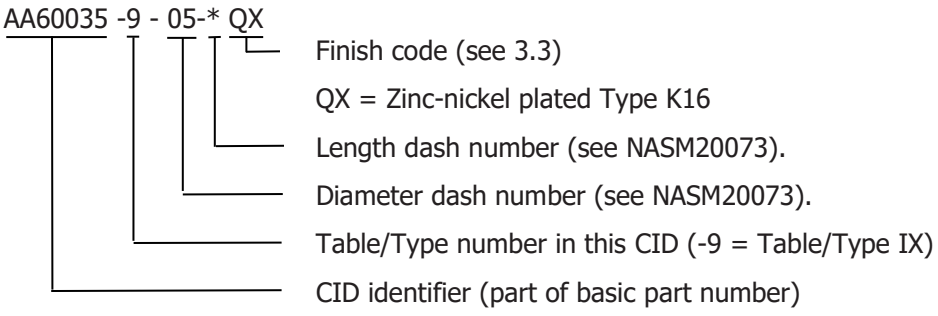
TABLE IX. Type IX (-9), bolt, machine, aircraft, drilled head, fine thread.

(Use NASM20073 for material, heat treatment (if any), and sizes/lengths.)

A-A-60035 Part Number Zinc-Nickel, Type K16 Plated (Clean Finish)	NASM20073 Reference Part/Dash Number Cadmium Plated, DO NOT USE on Ground Vehicles
AA60035-9-03-*QX	MS20073-03-*
AA60035-9-04-*QX	MS20073-04-*
AA60035-9-05-*QX	MS20073-05-*
AA60035-9-06-*QX	MS20073-06-*
AA60035-9-07-*QX	MS20073-07-*
AA60035-9-08-*QX	MS20073-08-*
AA60035-9-09-*QX	MS20073-09-*
AA60035-9-10-*QX	MS20073-10-*
AA60035-9-12-*QX	MS20073-12-*

NOTES:

PIN construction:



Example of a clean finish PIN:

AA60035-9-05-07QX = Bolt, 0.3125-24 UNF-3A thread, 0.375 grip, 0.922 long, zinc-nickel Type K16 plated.

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7.2 Source of documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/.

7.2.2 Executive order (EO). Copies of EO 14057 are available online from https://www.fedcenter.gov/programs/eo14057/.

7.2.3 Military specifications: Copies of military specifications are available online from https://quicksearch.dla.mil.

7.2.4 AIA documents: Copies of Aerospace Industries Association documents are available online from https://www.aia-aerospace.org/standards/.

7.2.5 ASTM documents: Copies of ASTM documents are available online from https://www.astm.org.

7.2.6 SAE documents: Copies of SAE documents are available online from https://www.sae.org.

7.3 Clean finish definition. On the tables, where it says clean finish, that is defined as treatments that do not include more than 0.1 percent by weight Cadmium or Hexavalent Chromium in any homogenous layer.

7.4 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this CID.
b. National stock number (if available).
c. Material (aluminum, CRES, steel alloy, or steel).
d. Hexavalent chromium and cadmium free part number (PIN) from this CID.
e. Quantity.

7.5 Key words:

Alloy steel
Aluminum
Carbon steel
Plating
Steel
Zinc-Nickel

7.6 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

Custodians:
Army – AT
Navy – AS
Air Force – 71
DLA – CC

Review activities:
Army – AR, AV, CR, MI
Navy – CG, MC, YD
Air Force – 03, 11, 70
DLA – GS4, GS7

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
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(Project 5306-2024-007)

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